

Dongxuan Zhu

dxzhu@se.cuhk.edu.hk | <https://dongxuan-zhu.github.io/>

Education

The Chinese University of Hong Kong, Ph.D. Aug. 2024 - Present

- Department of Systems Engineering and Engineering Management
- Supervised by Prof. Viet Anh Nguyen

Nanjing University, M.S. in Mathematics Sep. 2021 - Jun. 2024

- School of Mathematics
- Supervised by Prof. Houcai Shen & Prof. Caihua Chen

Sichuan University, B.S. in Mathematics and Applied Mathematics Sep. 2017 - Jun. 2021

- WUYUZHANG Honors College
- GPA: 3.75/4.0

Publications

D. Zhu, L. Tran, A. Cheung, M-C. Yue, and V. A. Nguyen, “Exploring diverse generation paths via inference-time Stiefel activation steering,” in International Conference on Learning Representations (ICLR), 2026.

L. Tran, D. Zhu, M-C. Yue, and V. A. Nguyen, “Test-time diverse reasoning by Riemannian activation steering,” in AAAI, 2026.

Preprints

D. Zhu, W. Huang, C. Chen, “Boosting accelerated proximal gradient method with adaptive sampling for stochastic composite optimization,” under review

Experience

Internship, Ant Group – Hangzhou, China Nov. 2023 - Mar. 2024

- Designed and implemented scalable first-order optimization methods for large-scale linear programming.
- Deployed optimization algorithms to production business workflows in finance and risk control.

Honors

- The First Prize Scholarship (2021) and The Third Prize Scholarship (2022) in Nanjing University;
- Individual Scholarship (2018&2019) and The Third Prize Scholarship (2020) in Sichuan University;
- Won Scholarship in honor of outstanding performance in CBL Summer Institute at Oriel Oxford & Cambridge Summer Institute in 2019;
- Awards for volunteer activities and students’ association activities.

Skills

Programming: Python, MATLAB

Technologies: Pytorch, Optimization solvers (e.g., Gurobi, IPOPT, MOSEK)

Languages: Mandarin (native), English (fluent)